

Arduino Alarm System

Final Project Documentation (with diagrams)

Controller: Arduino Micro (ATmega32U4)

Display: 1.44" TFT ST7735S (SPI)

Input: TP229 keypad, 4× photoelectric sensors (LS1–LS4)

Output: Siren + alarm LED

This document describes the system behavior, wiring, state machine and service procedures.

Version: 1.0

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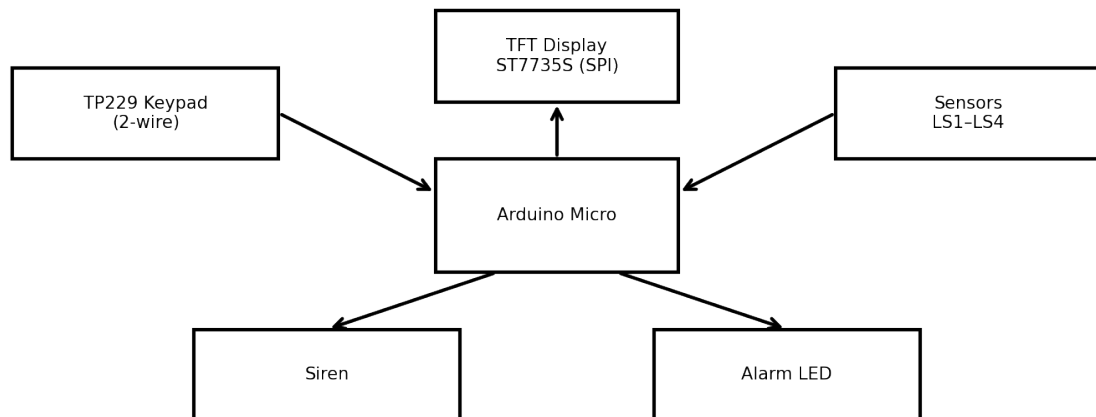
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1. Project Overview

The Arduino Alarm System is a code-protected alarm controller. When the system is armed and any sensor is triggered, the alarm is latched: the siren and alarm LED are activated and the TFT display flashes red until a valid code is entered to acknowledge the alarm. Acknowledgment stops the siren and automatically disarms the system.

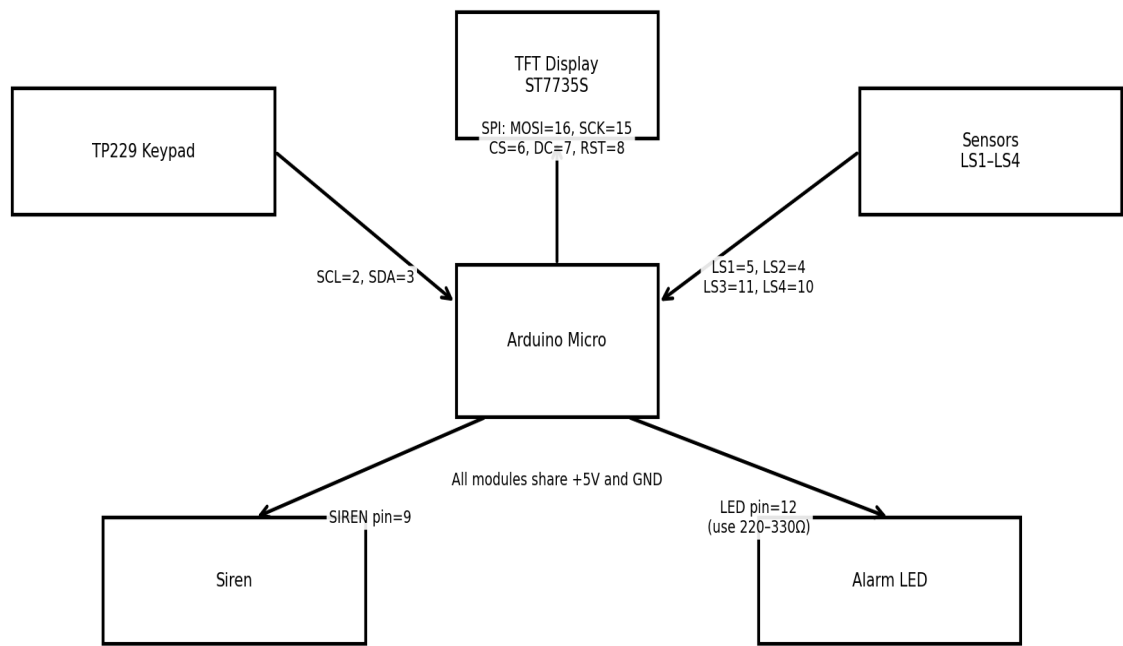
2. System Architecture

The following block diagram shows the major components and their connections.



3. Hardware & Wiring

Wiring overview (single Arduino Micro in the center). All modules share +5V and GND.



Pin Mapping

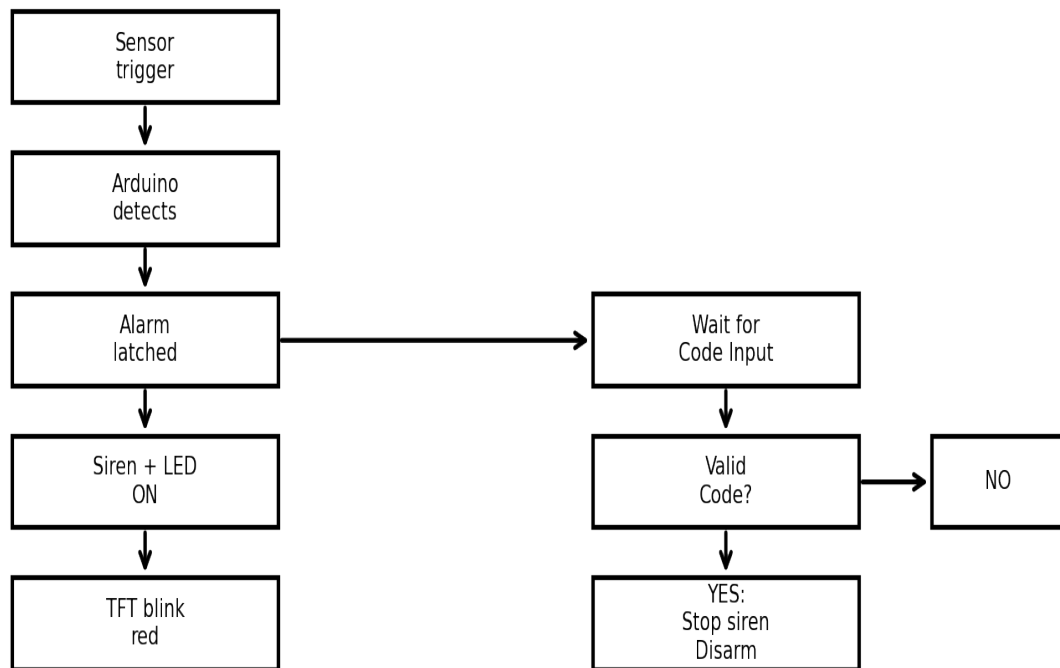
Subsystem	Signal	Arduino Micro Pin
TFT ST7735S	CS	6
TFT ST7735S	DC	7
TFT ST7735S	RST	8
TFT ST7735S	MOSI (SPI)	16
TFT ST7735S	SCK (SPI)	15
TP229 Keypad	SCL	2
TP229 Keypad	SDA	3
Sensors	LS1	5
Sensors	LS2	4
Sensors	LS3	11
Sensors	LS4	10
Outputs	Siren	9
Outputs	Alarm LED	12 (with 220–330Ω resistor)

4. Functional Behavior

- **Disarmed:** Sensors are ignored. Status display shows ALARM OFF.
- **Armed:** Sensors are monitored. Status display shows ALARM ON.
- **Alarm triggered:** When any sensor is triggered while armed, the alarm becomes latched.
- **Latched alarm:** Siren is ON, LED is ON, and the TFT flashes red/black.
- **Acknowledge:** Enter the valid 4-digit code and press #. The system stops the siren and disarms the alarm.

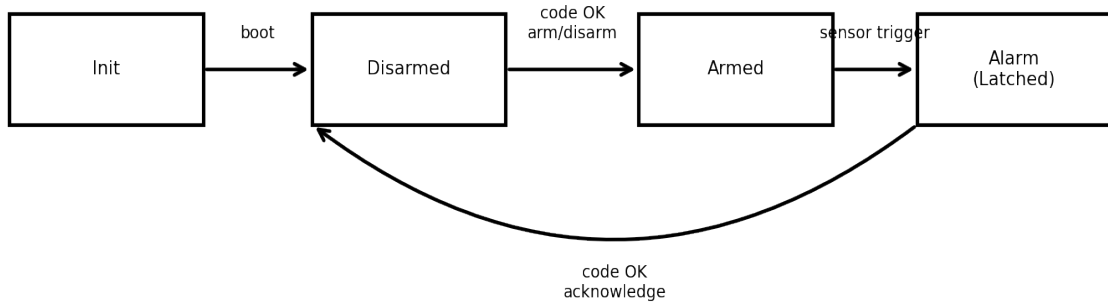
5. Signal Flow

The signal flow from sensor trigger to alarm acknowledgment is illustrated below.



6. State Machine

The controller logic is implemented as a finite state machine (FSM).



7. EEPROM Data

The system stores critical data in the internal EEPROM to survive power cycles:

Address	Length	Content
0	1 byte	Magic byte (initialized marker)
1–4	4 bytes	User code (4-digit)
5	1 byte	Alarm armed flag (0/1)

Note: EEPROM writes are performed using update semantics to avoid unnecessary wear.

8. Installation & Commissioning

1. Connect the TFT display and verify it powers on.
2. Connect the keypad and verify key presses are detected.
3. Connect the LED (with resistor) and siren.
4. Connect sensors LS1–LS4 (INPUT_PULLUP wiring).
5. Power on and set the initial 4-digit code.
6. Arm the alarm and test each sensor to trigger the alarm.
7. Acknowledge using the code (alarm should stop and disarm).

9. Maintenance & Troubleshooting

- **No display:** check VCC/GND, BL pin to 5V, SPI wiring (MOSI=16, SCK=15), CS/DC/RST pins.
- **Wrong colors/offset:** verify initializer INTR_144GREENTAB and rotation setting.
- **Sensors unstable:** confirm INPUT_PULLUP and correct sensor wiring to GND.
- **Siren does not stop:** ensure correct code entry and keypad wiring.

10. Bill of Materials (BOM)

Item	Qty
Arduino Micro	1
TFT 1.44" ST7735S	1
TP229 Keypad	1
Photoelectric sensors (light barriers)	4
Active siren (5V)	1
Alarm LED (red) + 220–330Ω resistor	1
Wires / connectors / enclosure (as required)	—